

# Library Database

## 1. Create Database and Collection

use Library1DB

db.createCollection("Books")

```
libraryDB> use Library1DB
switched to db Library1DB
Library1DB> db.createCollection("Books")
{ ok: 1 }
Library1DB> _
```

## 2. Insert Operations

Insert one document:

```
Library1DB> db.Books.insertOne({title:"The Great Gatsby",author:"F.Scott Fitzgerald",published_year:1925})
{
  acknowledged: true,
  insertedId: ObjectId('69f5f841d14f4d196a3682d4')
}
Library1DB> _
```

Insert multiple documents:

```
Library1DB> db.Books.insertMany([{title:"Harry Potter and the sorcerer's Stone",author:"J.K.Rowling",published_year:1997},
,{title:"The Catcher in the Rye",author:"J.D. Salinger",published_year:1951},{title:"To Kill a Mockingbird",author:"Harper Lee",published_year:1960},
,{title:"1984",author:"George Orwell",published_year:1949},{title:"MongoDB Basics",author:"John Doe",published_year:2022},{title:"NoSQL for Beginners",author:"Jane Smith",published_year:2021 }])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f5fb99d14f4d196a3682d5'),
    '1': ObjectId('69f5fb99d14f4d196a3682d6'),
    '2': ObjectId('69f5fb99d14f4d196a3682d7'),
    '3': ObjectId('69f5fb99d14f4d196a3682d8'),
    '4': ObjectId('69f5fb99d14f4d196a3682d9'),
    '5': ObjectId('69f5fb99d14f4d196a3682da')
  }
}
Library1DB>
```

## 3. Read Operations

Fetch all documents:

db.Books.find()

```
Library1DB> db.Books.find()
[
  {
    _id: ObjectId('69f5f841d14f4d196a3682d4'),
    title: 'The Great Gatsby',
    author: 'F.Scott Fitzgerald',
    published_year: 1925
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d5'),
    title: 'Harry Potter and the sorcerer's Stone',
    author: 'J.K.Rowling',
    published_year: 1997
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d6'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 1951
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d7'),
    title: 'To Kill a Mockingbird',
    author: 'Harper Lee',
    published year: 1960
  }
]
```

```

},
{
  _id: ObjectId('69f5fb99d14f4d196a3682d8'),
  title: '1984',
  author: 'George Orwell',
  published_year: 1949
},
{
  _id: ObjectId('69f5fb99d14f4d196a3682d9'),
  title: 'MongoDB Basics',
  author: 'John Doe',
  published_year: 2022
},
{
  _id: ObjectId('69f5fb99d14f4d196a3682da'),
  title: 'NoSQL for Beginners',
  author: 'Jane Smith',
  published_year: 2021
}
]
Library1DB>

```

Find and display only the documents where the author is "J.K. Rowling".

```

]
Library1DB> db.Books.find({author:"J.K.Rowling"})
[
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d5'),
    title: "Harry Potter and the sorcerer's Stone",
    author: 'J.K.Rowling',
    published_year: 1997
  }
]
Library1DB>

```

#### 4. Update Operations

Update one document:

Update the published year of the book with the title "The Catcher in the Rye" to the current year.

```

]
Library1DB> db.Books.updateOne({title:"The Catcher in the Rye"},{$set:{published_year:2024}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

```

Update multiple documents:

- Add a new field "genre" with the value "Mystery" to all documents in the "books" collection.

```

]
Library1DB> db.Books.updateMany({},{$set:{genre:"Mystery"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 7,
  modifiedCount: 7,
  upsertedCount: 0
}
Library1DB> db.Books.find()
[
  {
    _id: ObjectId('69f5f841d14f4d196a3682d4'),
    title: 'The Great Gatsby',
    author: 'F.Scott Fitzgerald',
    published_year: 1925,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d5'),
    title: "Harry Potter and the sorcerer's Stone",
    author: 'J.K.Rowling',
    published_year: 1997,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d6'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 2024,

```

```

    genere: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d7'),
    title: 'To Kill a Mockingbird',
    author: 'Harper Lee',
    published_year: 1960,
    genere: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d8'),
    title: '1984',
    author: 'George Orwell',
    published_year: 1949,
    genere: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d9'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genere: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682da'),
    title: 'NoSQL for Beginners',
    author: 'Jane Smith',
    published_year: 2021,
    genere: 'Mystery'
  }
}

```

## 5. Delete Operations

Delete one document:

- Remove the document with the title "1984" from the "books" collection.

```

Library1DB> db.Books.deleteOne({title:"1984"})
{ acknowledged: true, deletedCount: 1 }

```

Delete all documents:

- Delete all documents from the "books" collection where the published year is before 2000.

```

{ acknowledged: true, deletedCount: 3 }
Library1DB> db.Books.deleteMany({published_year:{$lt:2000}})
{ acknowledged: true, deletedCount: 3 }
Library1DB>

```

## 6. Query Operators

- Find and display the top 3 recently published books from the "books" collection.

- Retrieve documents from the "books" collection where the title contains the word "MongoDB" or "NoSQL".

Remember to use the appropriate MongoDB shell commands (e.g., `use`, `db.collection.insert`, `db.collection.find`, `db.collection.update`, `db.collection.remove`, etc.) to perform these operations. This will help you become more familiar with MongoDB's CRUD capabilities.

Type:1

```

{ acknowledged: true, deletedCount: 3 }
Library1DB> db.Books.find().sort({published_year:-1}).limit(3)
[
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d6'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 2024,
    genere: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d9'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genere: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682da'),
    title: 'NoSQL for Beginners',
    author: 'Jane Smith',
    published_year: 2021,
    genere: 'Mystery'
  }
]
Library1DB>

```

Type:2

```
Library1DB> db.Books.aggregate([{$set:{published_year:-1}},{$limit:3}])
[
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d6'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: -1,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d9'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: -1,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682da'),
    title: 'NoSQL for Beginners',
    author: 'Jane Smith',
    published_year: -1,
    genre: 'Mystery'
  }
]
Library1DB>
```

Type:3

```
Library1DB> db.Books.find({}, {title:1,published_year:1,_id:0}).sort({published_year:-1}).limit(3)
[
  { title: 'The Catcher in the Rye', published_year: 2024 },
  { title: 'MongoDB Basics', published_year: 2022 },
  { title: 'NoSQL for Beginners', published_year: 2021 }
]
Library1DB>
```

Retrieve Documents with title MongoDB & NoSQL

Type:1

```
Library1DB> db.Books.find({$or:[{title:/MongoDB/i},{title:/NoSQL/i}]})
[
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d9'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682da'),
    title: 'NoSQL for Beginners',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
Library1DB>
```

Type:2

```
Library1DB> db.Books.find({title:{$in:[/MongoDB/i, /NoSQL/i]}})
[
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d9'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682da'),
    title: 'NoSQL for Beginners',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
Library1DB> 
```

Type:3

```
]
Library1DB> db.Books.aggregate([{$match:{title:{$regex:"MongoDB|NoSQL", $options:"i"}}}])
[
  {
    _id: ObjectId('69f5fb99d14f4d196a3682d9'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f5fb99d14f4d196a3682da'),
    title: 'NoSQL for Beginners',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
Library1DB>
```